A Statistical Case for (Qualified) Scientific Optimism

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based on the paper with Erik van Zwet and Andrew Gelman

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AI disclaimer

Some of the *presentation* text was generated by an LLM (last night). The paper is 99% an old-fashioned artisanal product.

All mistakes are LLM's.

Studying the quality of empirical science

- The replication crisis refuted the intuition that a significance threshold can tell us whether a study is good.
- Even $p < 0.01$ versus $p > 0.2$ does not discriminate well between high and low signal-to-noise ratios (SNR = effect size / SE).
- Many people remain dismayed when results “fail to replicate”, despite knowing that we should not rely on a single study.
- Can we reconcile these problems with a more positive view of statistically based science?

Two-sided p-values of 0.01 and 0.2 correspond to $|z| = 2.58$ and 1.28. These are about equally likely at a signal-to-noise ratio of $(2.58 + 1.28)/2 = 1.93$, which corresponds to 49% power.

Ioannidis 2005: binary model of scientific findings

Research findings are compared against “true relationships” and “no relationships” in a scientific field.

	Ground Truth	
	No relationship	True relationship
Significant	False positive	True positive
Non-significant	True negative	False negative

Null studies:	$\text{SNR} = 0$	$\text{PoS} = 0.05$
Non-null studies, e.g.:	$\lvert\text{SNR}\rvert = 2.8$	$\text{PoS} \approx 0.80$

- Low TR/NR ratio = most published findings are false = low replication rate

Ioannidis (2005). *Why Most Published Research Findings Are False.* PLoS Medicine 2(8): e124.

Limitations of the 2×2 model

1. The goal is not to separate "no relationships" from "true relationships".

Effects of interest are rarely exactly zero. With finite data, it is not possible to distinguish effects that are zero from those that are small.

2. The binary model is not a realistic representation of empirical research.

In particular, assumption of some fixed power for TR's. Researchers do not know the true effect when designing a study. SNRs are on a continuum.

Two contributions of this paper

Data: Benchmarks of Empirical Accuracy in Research (BEAR)

- We make a large number of results from empirical research easily accessible.
- We process the available data to make them ready for metascientific research, with clear documentation and open-source code.

Method: a model of empirical z-values and signal-to-noise ratios

- Extend existing models to accommodate selective reporting and truncation.
- Estimate main metascientific characteristics of fields of research (power, sign, replication).

Stephens (2017), *False discovery rates: a new deal*; van Zwet et al. (2021), *The statistical properties of RCTs and a proposal for shrinkage*; van Zwet and Gelman (2022), *A proposal for informative default priors scaled by the standard error of estimates.*

BEAR

Benchmarks of Empirical Accuracy in Research

24 datasets · 12.5 million results · 7,476 meta-analyses

- Existing meta-analyses, systematic reviews, trial registries, metascience datasets and replication projects.
  - Side note: better databases of *replications* elsewhere.
- This paper: 16 corpora, 374,340 results from 183,795 studies.
- Medicine, economics, psychology, psychotherapy, ecology and evolution, political science, education, and other fields.
- We do not contribute any new data; all data collection and the vast majority of processing were done by the creators of the source datasets.

Accessing BEAR

Website and documentation
https://witold.xyz/BEAR/

Download the combined data
github.com/wwiecek/BEAR/releases

Download `BEAR.rds`, then load it in R:

```
bear <- readRDS("BEAR.rds")
```

Code and individual datasets
github.com/wwiecek/BEAR · github.com/wwiecek/BEAR_data

What does one observation look like?

Corpus	Study ID	Measure	Estimate	SE	z	Year
Cochrane	Dwyer 2015	SMD	-0.160	0.330	-0.49	2015
Cochrane	Dwyer 2015	SMD	0.122	0.330	0.37	2015
Cochrane	Dwyer 2015	SMD	0.675	0.339	1.99	2015
Nuijten	32	Unspecified	0.371	0.052	7.19	2003
BarnettWren	12814712	ratio	-0.186	0.129	-1.45	2003
BarnettWren	29794831	ratio	0.824	0.358	2.30	2018

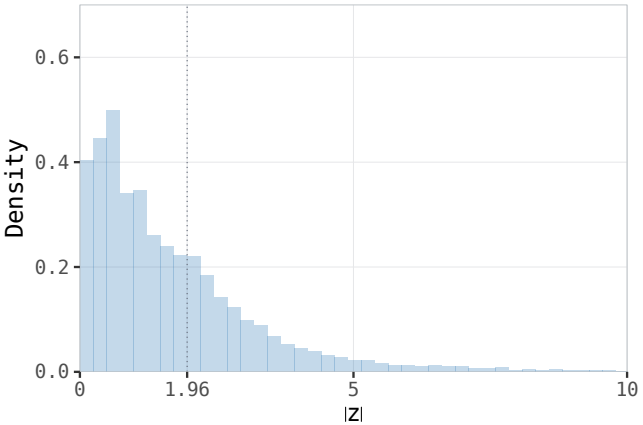
- The first three rows are **three results from one study**, Dwyer (2015).
- A study identifier is interpreted within its corpus.
- In most cases we have one or several estimates per study.
- Estimates and SEs retain their BEAR scales.
- Many datasets contain rich additional study characteristics.

SMD: standardised mean difference. “Unspecified” denotes missing measure metadata. Ratio estimates and SEs are on the log scale.

Corpora used in this paper

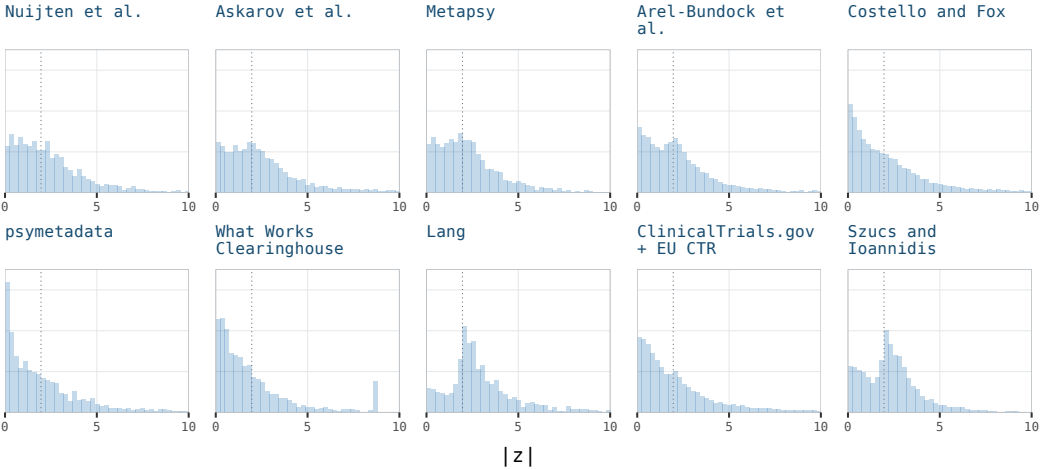
Corpus	Domain	N studies	Results/study	% signif.
Curated sets				
Nuijten et al.	intelligence	1,913	1.3	53%
Askarov et al.	economics	1,913	11.2	52%
Metapsy	psychotherapy	1,494	2.9	48%
Arel-Bundock et al.	political science	2,252	7.4	47%
Costello and Fox	ecology & evolution	12,927	6.8	42%
psymetadata	psychology	721	11.8	36%
What Works Clearinghouse	education	1,402	5.0	34%
Cochrane	medicine & health	26,095	1.3	33%
Representative domain samples				
SCORE claims	social / behavioural	159	12.2	87%
Chavalarias et al.	biomedicine	50,000	1.0	87%
Barnett and Wren	biomedicine	50,000	1.0	83%
Szucs and Ioannidis	cognitive neuroscience	2,261	7.5	62%
Lang	economics	730	5.3	58%
ClinicalTrials.gov + EU CTR	clinical trials	31,668	2.2	44%
Replication efforts				
SCORE replications	social / behavioural	163	1.6	64%
Open Science Collaboration	psychology	97	1.0	35%

Cochrane Database of Systematic Reviews



26,095 studies
33,507 results
33% significant
Most mass lies near zero, with a long right tail.
The dotted line marks $|z| = 1.96$.

Distribution of absolute z-values across other corpora



z-values and signal-to-noise ratios

For study i, true effect β_i and estimate $\widehat{\beta}_i$:

$$z_i = \frac{\widehat{\beta}_i}{\mathsf{SE}(\widehat{\beta}_i)}, \qquad \mathsf{SNR}_i = \frac{\beta_i}{\mathsf{SE}(\widehat{\beta}_i)}.$$

Suppose that we have unbiased effect estimates which are normally distributed with known standard errors. Then

$$z_i = \mathsf{SNR}_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0,1),$$

where the error is independent of the SNR.

For two-sided p-values:

$$p = 0.05 \Rightarrow |z| = 1.96, \quad p = 0.01 \Rightarrow |z| = 2.58, \quad p = 0.001 \Rightarrow |z| = 3.29.$$

A model for the distribution of z-values

Our main assumption is that the distribution of the absolute z-values is well represented by a **mixture of half-normal distributions**.

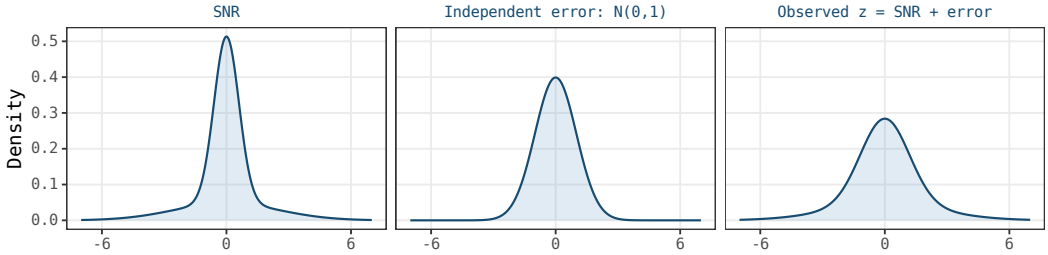
Equivalently, we model signed z-values as a mixture of zero-mean normals:

$$f(z \mid p, \sigma) = \sum_{i=1}^{k} p_i \frac{1}{\sigma_i} \phi\left(\frac{z}{\sigma_i}\right).$$

- ϕ is the standard normal density.
- $p = (p_1, \ldots, p_k)$: non-negative mixture weights that sum to unity.
- $\sigma = (\sigma_1, \ldots, \sigma_k)$: standard deviations, at least 1 because every z-value has a standard normal noise component.
- We use $k = 4$. No need to go higher, could go lower in most cases.

From signal-to-noise ratios to z-values

The z-value is equal to the SNR plus an independent standard normal error.



Deconvolution: the distribution of the SNRs

The distribution of the z-values is a convolution of the distribution of the SNRs and the standard normal error:

$$z = \mathsf{SNR} + \varepsilon, \qquad \varepsilon \sim N(0, 1).$$

We simply **subtract 1 from the variances of the mixture components**:

$$f_{\mathsf{SNR}}(x \mid p, \sigma) = f\left(x \mid p, \sqrt{\sigma^2 - 1}\right).$$

Component i has variance σ_i^2 for z and $\sigma_i^2 - 1$ for SNR.

Adjustment for selective reporting

Following Hedges, define ω as the relative risk that a result with $|z| < 1.96$ is observed compared to a result with $|z| \geq 1.96$:

$$\omega = \frac{\mathsf{Pr}(\text{result is published} \mid |z| < 1.96)}{\mathsf{Pr}(\text{result is published} \mid |z| \geq 1.96)}.$$

The density conditional on publication becomes

$$f(z \mid p, \sigma, \omega, \mathsf{pub}) = \begin{cases} \omega\, f(z \mid p, \sigma)/C(p, \sigma, \omega), & |z| < 1.96, \\ f(z \mid p, \sigma)/C(p, \sigma, \omega), & |z| \geq 1.96. \end{cases}$$

The normalising constant is

$$C(p, \sigma, \omega) = \omega\, \mathsf{Pr}(|z| < 1.96 \mid p, \sigma) + \mathsf{Pr}(|z| \geq 1.96 \mid p, \sigma).$$

Hedges (1984, 1992). "pub" denotes conditioning on publication.

Truncation, repeated estimates, and likelihood

Reported values are sometimes truncated.

- Statements such as $p < 0.05$, $p < 0.01$, or $p < 0.001$ give bounds on $|z|$.
- $\tilde{z}$ denotes the possibly censored value; δ indicates left censoring, right censoring, or no censoring.

Most datasets have multiple z-values per study.

- To prevent studies with many z-values from dominating the likelihood, weight observations by the inverse number of z-values per study.

For n observations, we maximise the weighted log likelihood

$$\ell(p, \sigma, \omega) = \sum_{j=1}^{n} w_j \log f(\tilde{z}_j, \delta_j \mid p, \sigma, \omega, \text{pub}),$$

Here w_j is the weight of observation j.

Quantities implied by the distribution of the SNRs

Probability of significance

$$\mathrm{PoS}(\mathrm{SNR}) = \mathrm{PoS}(|\mathrm{SNR}|) = \Pr(|z| \geq 1.96 \mid \mathrm{SNR}).$$

Averaging over the distribution of the SNR gives the **assurance**, $\overline{\mathrm{PoS}}$.

Conditional probability of the correct sign

$$\Pr(z \cdot \mathrm{SNR} > 0 \mid |z|).$$

Conditional probability of "successful replication"

$$\Pr(z_{\mathrm{repl}} \cdot z > 0 \text{ and } |z_{\mathrm{repl}}| > 1.96 \mid |z|).$$

We imagine a second study with **exactly the same SNR**: $z_{\mathrm{repl}} = \mathrm{SNR} + \varepsilon_{\mathrm{repl}}$, with independent $\varepsilon_{\mathrm{repl}} \sim N(0, 1)$.

Conditional inference for SNR given z

The marginal distribution of z, together with the conditional distribution of z given SNR, determines the joint distribution.

The conditional distribution of **SNR given** z is a mixture of normals:

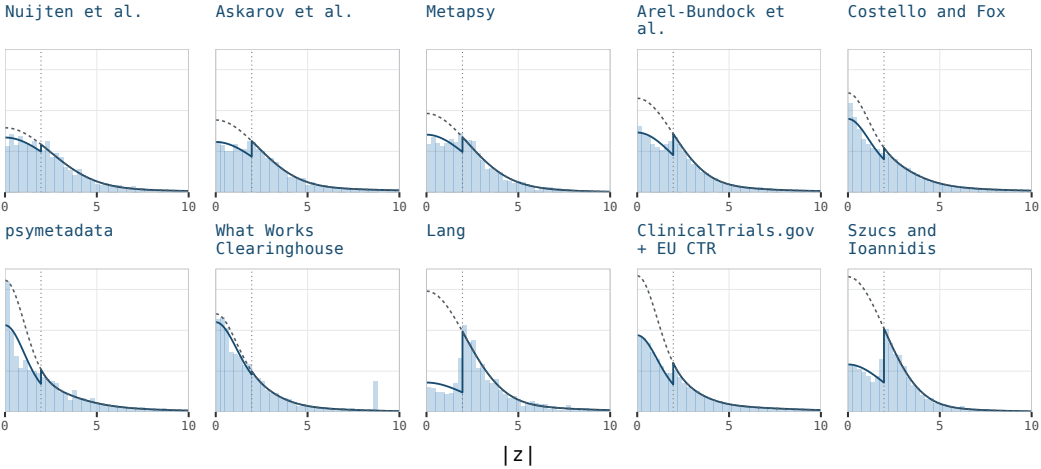
$$\text{SNR} \mid z \sim \sum_{i=1}^{k} q_i(z)\, N(\mu_i(z), \tau_i^2)\,,$$

with weights, means, and variances

$$q_i(z) = \frac{p_i \phi(z/\sigma_i)/\sigma_i}{f(z \mid p, \sigma)}, \qquad \mu_i(z) = \frac{\sigma_i^2 - 1}{\sigma_i^2}\, z, \qquad \tau_i^2 = \frac{\sigma_i^2 - 1}{\sigma_i^2}.$$

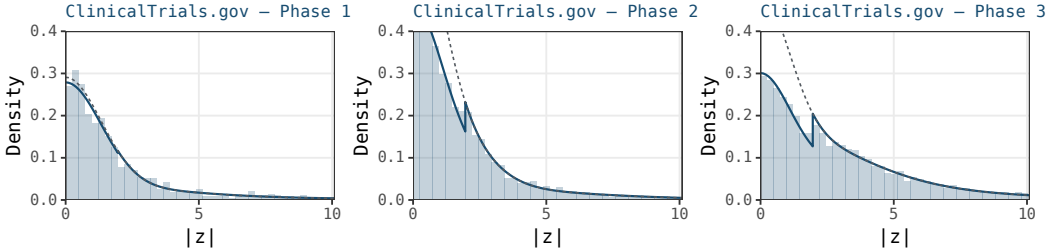
For an exact replication, $z_{\text{repl}} \mid z$ has the same weights and means, with variances $\tau_i^2 + 1$.

Model fit across corpora



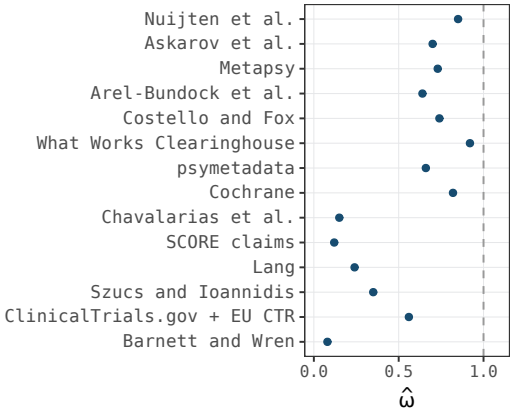
Solid: fitted density conditional on publication. Dashed: corrected density, rescaled above $|z| = 1.96$.

Decreasing distributions of absolute z-values



- Mixtures of half-normal densities are necessarily decreasing. But data support this strongly.
- The main difference between the distributions of the z-values is their **widths**. A wider distribution of z-values implies a wider distribution of SNRs.
- Since PoS is an increasing function of the absolute SNR, a higher proportion of results will reach statistical significance.

What does the selection parameter tell us?

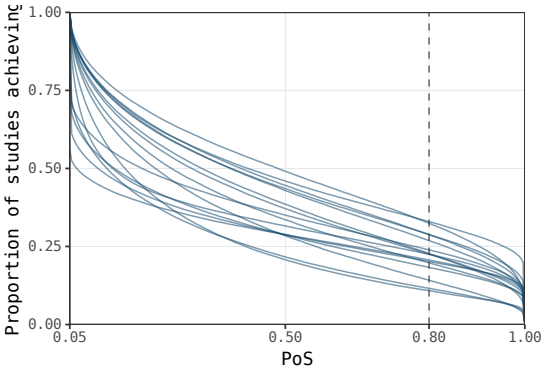


$\hat{\omega}$ ranges from **0.08 to 0.92**.

- Many studies have multiple outcomes; we do not know which drive publication decisions.
- Hedges' single-parameter model is useful for **adjusting** the distributions of z-statistics.
- It seems too limited for reliably quantifying and comparing the extent of publication bias across fields.

Paper Table 2. Replication datasets removed ($\omega = 1$ under no selection).

Distribution of the probability of significance

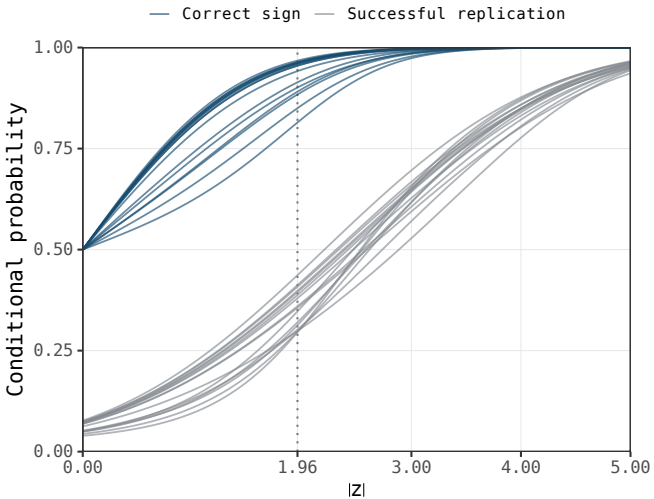


Estimated assurance: **28–52%**.
Only **11–33%** of studies achieve 80% PoS.

This does not tell us if studies are too small!
PoS is “power”, but what matters is power to detect a meaningful effect.

Each curve represents a corpus. Paper Tables 2–3; adapted from the PoS figure. Senn (2002).

Correct sign versus successful replication



Unconditionally:

- **72–89%** probability of the correct sign.
- **26–50%** probability of "successful replication".

At $|z| = 1.96$:

- **81–97%** probability of the correct sign.
- **30–44%** probability of "successful replication".

Paper Table 2 and conditional-probability figure. Replication has exactly the same SNR and independent sampling error.

Marginal significance and the base rate of replication

If $|z| = 1.96$, the probability of "successful replication" is never much more than the base rate, and often even less.

Corpus	Randomly selected study	Conditional on $\|z\| = 1.96$
Nuijten et al.	47%	41%
Askarov et al.	46%	40%
Cochrane	26%	30%
Chavalarias et al.	50%	44%

- This phenomenon is **not a consequence of many studies having low SNRs** (low PoS).
- If all the studies had higher SNRs, the assurance would be higher, but observing $|z| = 1.96$ would be even less impressive.

Paper Table 2. Both columns report the probability of a significant, same-direction exact replication.

Results summary

- The estimated distributions generally track the histograms well.
- The main difference between the distributions of the z-values is their widths; discontinuities at $|z| = 1.96$ are a feature of the selection model.
- Estimated assurances range from **28–52%**; **11–33%** of studies achieve 80% PoS.
- For a marginally significant result, the probability of "successful replication" is **30–44%**.
- The probability that the direction of the observed effect is correct is **81–97%**.

Paper Tables 2–3. The last two statements condition on $|z| = 1.96$.

Four features of the empirical corpora

1. **Absolute z-values have a decreasing density.**
   Studies with low SNRs are more common than those with high SNRs. Research domains differ considerably in the widths of these distributions.
2. **The probability of significance tends to be low.**
   Only 11–33% of studies achieve 80% power.
3. **Reaching statistical significance does not imply a high probability of replication.**
   At $|z| = 1.96$, the probability of successful replication is 30–44%.
4. **Statistical significance does imply that the sign is likely correct.**
   At $|z| = 1.96$, the probability of the correct sign is 81–97%.

Interpretation: qualified scientific optimism

- Our model allows the PoS to range continuously from 0.05 to 1. We observe no clear separation between “true relationships” and “no relationships”.
- Empirical research often leads to true findings—in the sense of correctly identifying the direction of an effect—that do not replicate in the sense of achieving statistical significance in the same direction.
- Failure to replicate should not be taken to mean that the original result was a fluke or a fake.

Our qualified optimism extends only to research for which we do not have decisive reasons to distrust.

Which metascience claims survive the empirical data?

An opinionated summary

Supported by this analysis

- Significant results often fail to replicate significantly.
- Most studies have modest true PoS; only 11–33% achieve 80% power. This alone does not establish poor study planning.

Need qualification or reframing

- "Most published results are false"; some fields are bad and others "gold standard".
- Lowering the p-value threshold or uniformly doubling sample sizes does not follow from these results.

Not answered by this paper

- How common publication bias and p-hacking are.
- How much research funding should go to replication.